N9 formal proof-boundary report

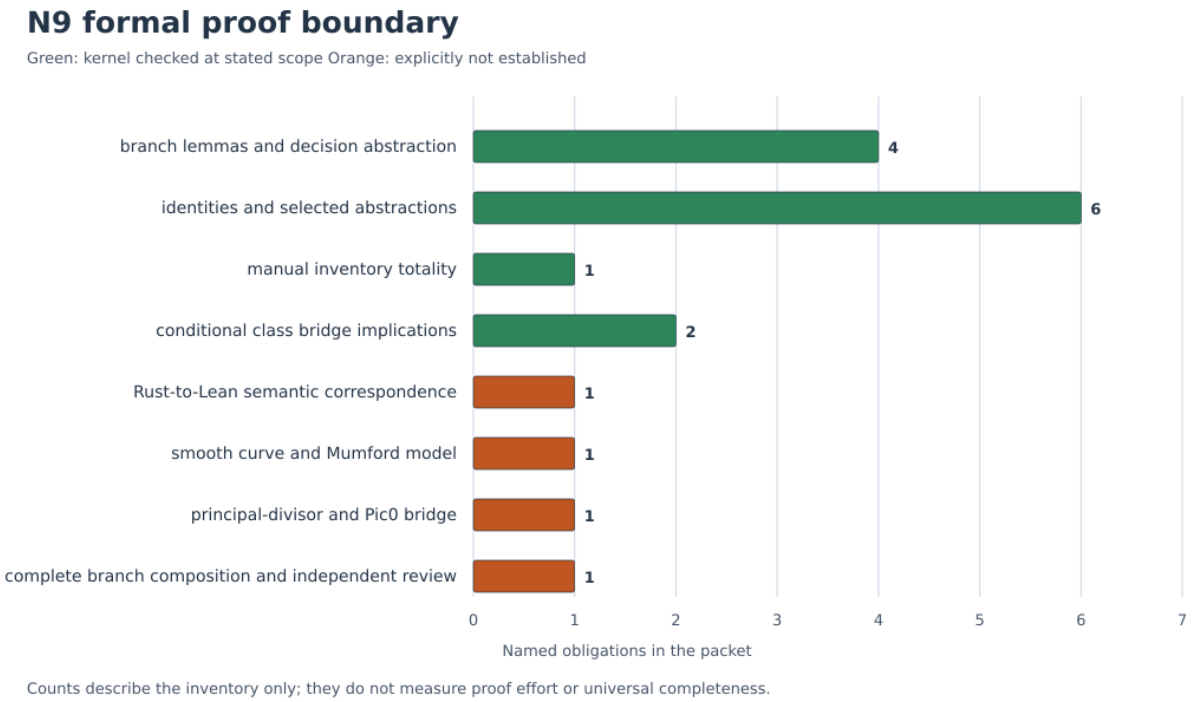
Partial mechanization with executable abstractions • generated 2026-10-25

Schema: `projective-law-formal-report/v1`

Recorded result

Lean 4.34.1 with pinned Mathlib revision `d13f23b723b8a846827a245b89c10fc7d3f11612` kernel-checks the prior branch and identity boundary plus a manual executable abstraction of the 1+2 decision order, selected 1+2 intermediates, a proof-side 2+2 witness, and the abstract principal-divisor residual-class step.

Four exact Rust function fragments are separately SHA-256 bound and re-extracted by the verifier. They identify source bytes associated with the manual abstractions; they do not prove a translation. The packet does not establish Rust-to-Lean semantic equivalence, a smooth genus-2 curve or Mumford representation in Lean, the principal-divisor and Picard-group bridge, complete exceptional-branch composition, or independent mathematical review. The conditional bridge theorem constructs no bridge and assumes no bridge globally.



Counts are an inventory of named obligations. They do not measure proof effort, importance, or universal coverage.

Standardized source table

Evidence	Status	Scope or limit
Three branch lemmas	Kernel checked	Abstract-field implications with explicit hypotheses.
Generic 1+2 identity	Kernel checked	Literal commutative-ring identity; no divisor argument.
Two generic 2+2 identities	Kernel checked	Exact factorizations by explicit input quadratics.
Executable abstractions	Kernel checked, conditional	Decision order and selected intermediates under supplied hypotheses.
2+2 proof witness	Kernel checked	Constructed congruence witness; no Rust variable or source derivation is asserted.
Residual class theorem	Kernel checked, conditional	Final group step through an explicit interface; no concrete curve instance.
Runtime mapping	Kernel checked, manual scope	Total over a manually transcribed 17-row inventory; Rust is hash-bound, not parsed by Lean.
Conditional bridge	Kernel checked, conditional	Applies a caller-supplied implication; supplies no geometric bridge.
Source bindings	Integrity checked	Four function fragments, runtime, audit, and eight certificate pairs are SHA-256 bound.

Validation and remaining boundary

The pinned rebuild completed 742 jobs. The axiom audit found only subsets of `propext`, `Classical.choice`, and `Quot.sound`. The packet verifier and SHA-256 manifest passed. Live negative verifier tests rejected a duplicate binding, a hash-valid unexpected Rust function, an unexpected real Lean declaration, and a name present only in a Lean comment.

The next obligation is a proved Rust translation plus a concrete smooth-curve and Mumford model satisfying the principal-divisor interface for every branch, including exceptional compositions and degree drops. Until those bridges are supplied and independently reviewed, this packet records algebraic sublemmas and manual executable abstractions rather than a universal group-law proof.